

QUEUE

Q3)

Question:

DS	Session	Queue	Question Information	Level 1	Challenge 53
Question description Ramesh is an DS expert training youngsters struggling in DS to make them better. Ramesh usually gives interesting problems to the youngsters to make them love the DS. One such day Sathya provided to the youngsters to solve the task such that, delete an element in a Queue, Queue data structures work on the FIFO architecture so the element that has entered first in the list will go out from the list first. Youngsters were lacking the idea to solve the problem. Being an exciting youngster can you solve it?					

PROGRAM:

```
1  #include <stdio.h>
2
3  int main() {
4      int n, i;
5      int q[100];
6      int front = 0, rear = -1;
7
8      scanf("%d", &n);
9
10     for (i = 0; i < n; i++) {
11         scanf("%d", &q[++rear]);
12     }
13
14     printf("Dequeuing elements:\n");
15
16     while (front <= rear) {
17         front++;
18
19         for (i = front; i <= rear; i++) {
20             printf("%d ", q[i]);
21         }
22
23         printf("\n");
24     }
25
26     return 0;
27 }
```

OUTPUT:

TEST CASE 1:

```
5
5 6 3 1 2
Dequeuing elements:
6 3 1 2
3 1 2
1 2
2
```

TEST CASE 2:

```
8
1 8 6 4 9 5 3 5
Dequeuing elements:
8 6 4 9 5 3 5
6 4 9 5 3 5
4 9 5 3 5
9 5 3 5
5 3 5
3 5
5
```